ÖZYEĞİN UNIVERSITY

Course Syllabus
MATH 112 Discrete Mathematics
Spring 2025-26

INSTRUCTOR	
Name	Selçuk Demir
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Phone Number	-
Office Number	AB1.G01H
Office Hours	By e-mail
Research/Teaching Assistants	-

COURSE ORGANIZATION	
Interaction Methods	LMS will be used for announcements and sharing course material.
Technology Used by the Instructor	LMS, Meet, Zoom
Technology Requirements for Students	*A smartphone to take attendance*

COURSE MATERIALS	
Required Books	-
Recommended Books	- Discrete Mathematics and its Applications, by K. H. Rosen, published by McGraw-Hill, 8th edition, 2018. - Discrete and Combinatorial Mathematics, by R. P. Grimaldi, published by Pearson, 5th edition, 2017.
Other Materials	-
Accessing Course Materials	There will be worksheet problems assigned from Rosen's book. They will not be graded and the students are not expected to return the solutions but they are strongly advised to solve them.

WEEKLY SUBJECTS (TENTATIVE)		
Week	**Date**	**Subject**
1	09-13.02.2026	Logic and Proofs: Propositional logic
2	16-20.02.2026	Logic and Proofs: Predicate logic, Introduction to proofs
3	23-27.02.2026	Logic and Proofs: Proof methods, Mathematical induction
4	02-06.03.2026	Sets and Functions, Counting: Elementary principles
5	09-13.03.2026	Counting: Standard models (permutations, combinations, multisets), Applications and Binomial coefficients
6	16-20.03.2026	Counting Exercises
7	23-27.03.2026	Pigeonhole principle, Inclusion-exclusion principle
8	30.03-.03.04.2026	Advanced counting: Recurrence relations
9	06-10.04.2026	Advanced counting: Generating functions
10	13-17.04.2026 ***(Withdrawal Week)***	Probability: Discrete probability, Discrete random variables
11	20-24.04.2026	Probability: Discrete random variables
12	27.04-01.05.2026	Graph Theory: Introduction, Isomorphisms, Connectivity, Euler walks
13	04-08.05.2026	Graph Theory: Trees, Graph colorings
14	11-15.04.2026	Graph Theory: Examples

ASSESSMENT METHODS, WEIGHTS AND RULES			
Type	**Weight**	**Implementation Rule**	**Makeup Rule**
Final Exam	*40%.*	-Exams are closed book, closed notes -No computing or communication devices are allowed during the exams. -More details will be announced on LMS in due time.	BÜT exam will serve as the makeup for the final exam.
Midterm Exam (2 Midterm Exams}	*60%* *(30+30%)*	- There will be **two midterm exams.** - Exams are closed book, closed notes. - No computing or communication devices are allowed during the exams. - More details will be announced on LMS in due time.	- Each student can make up **only one of the midterm exams** due to sick leave in the semester. - Students must submit a health report to the department secretary by the end of the third business day after the date of the midterm exam(s) they missed. For details, see Article 28 - (3) of OzU Rules and Regulations. - The make-up for the midterm exams will be given at a date towards the end of classes. - There is no make-up for the make-up exam.
Attendance/ Interaction (Participation)	*Bonus points if applicable. See below for the rules.*	See below for the rules.	There will be **no make-ups or excuses** to attendance records.
Total	**100%**		

METHODS FOR TRACKING ATTENDANCE & IMPACT OF ATTENDANCE ON SUCCESS			
Method	**Application Details**	**Required Tools and Make-up Method**	**Impact of Attendance of Course Grade**
QR Code (via LMS) (or MyOZU)	During class, a QR code will be displayed on the screen. Scanning this code with the MyOzU app on your mobile device will be accepted as proof of attendance.	You must have a smartphone with the MyOzU app installed. If you do not have such a device, inform the instructor during class and sign the attendance sheet.	- Attendance will be taken at random throughout the semester and may occur more than once per day. - Each time attendance is taken counts as one attendance check. - There will be no attendance check before the add-drop period and at the last course day of the semester. - At least 70% must be attended. - Students who attend all sessions will receive 3 **bonus points**. - Students who are not present for at least 70% of all attendance checks will get 0 bonus points. - For the students who are present for at least 70% of the attendance checks, the bonus points will be calculated as follows: **Bonus attendance point**=3*(# of presence in att. checks / total # of att. checks).

ASSESSMENT(S) BEFORE THE WITHDRAWAL DEADLINE		
Type of Assessment(s)	**Weight(s)**	**Date by When the Score of the Assessment is to be Announced**
Midterm 1	30%	**No later than 15.04.2026**

DETERMINING LETTER GRADE

A student's course evaluation is based on the scores of 2 midterm exams and 1 final exam.

Student's total score is calculated according to the formula.

Total score = 0.30 x MT1 + 0.30 MT2 + 0.40 x FINAL + Bonus Attendance points

where the inputs are out of 100 points. If the total score of a student is greater than 100, it is going to be considered as 100. Letter grades will then be determined by the total class average and the distribution of the total scores.

To get a passing grade from this course, a student must have

1. Final Exam score of at least 10 points, and
2. a total score of at least 40 points.

Note that since BÜT exam will also serve as the makeup for the final exam, students taking the BÜT exam must also get a score of at least 10 points from the composite exam and a total score of at least 40 points to get a passing grade from this course.

EXPECTED STUDENT SEMESTER WORKLOAD	
Activities Carried Out in The Presence of An Instructor	**Total Hours in Semester**
Lecture (Including Discussion Sessions)	42
Activities Carried Out by The Learner Him/Herself	**Total Hours in Semester**
Working on Assignments	56
Pre-class Learning of Course Material	14
Review of Course Material for Exam Preparation	38
Overall Total Hours in Semester	**150**

RULES FOR THE USE OF AI IN THE COURSE	
Use of AI in the Course **(Not Allowed)**	The use of AI tools (ChatGPT, Copilot, Gemini, Claude, Grok, DeepSeek, etc.) is **not permitted** in this course. Assignments, projects, reports, etc. must be prepared solely with your own original contribution. The use of AI will be considered a violation of academic integrity. Note: If you use AI for work outside of this course, please keep in mind that you are expected to comply with the **Principles for the Responsible Use of AI** listed below as well as university policies.

OTHER RULES AND INFORMATION	
Health	The mode of instruction is subject to change in case of pandemic or other similar unavoidable reasons in line with the directives of the authorized public bodies. In such a case, the necessary revisions will be binding for all members of the OzU community, and will be announced by the relevant units, in particular Student Services.
Safety	Announced safety procedures must be strictly followed during any laboratory and similar work. Please refer to the safety guidelines posted in the laboratories. In case of emergencies, call 9911 (216 564 9911).
Accessibility	Reasonable accommodations will be provided for students with verifiable disabilities. Please inform your instructor if you need any assistance.
Academic and Technical Support	Students can get help from the Writing Center, Solution Center and IT in case they need academic and/or technical support with their classes.
Objections to Examination Results	Students who object to their exam results and cannot resolve their objections with the instructor of the course, may file a written petition for a review of their exam papers at their Dean's/Director's Office within five business days from the announcement their exam results.
Academic Integrity and Honor Code	Students are assumed to abide by scholastic honesty and uphold the Honor Code. Scholastic dishonesty includes, but is not limited to cheating, attempting to cheat, plagiarizing, fabricating information or citations, facilitating acts of dishonesty by others, having unauthorized possession of examinations, submitting the work of another person, or tampering with the academic work of other students. Any form of scholastic dishonesty is a serious academic violation and will result in a disciplinary action and also may result in student getting a 0 grade in the related assessment.
Unauthorized Recording/Sharing of Lectures	Recording/sharing of synchronous/asynchronous lectures or copying of lecture recordings are prohibited without the permission of the instructor. Actions such as unauthorized recording, giving the recording to someone else, taking it, using it or sharing it in physical or virtual media are prohibited and/or are offenses according to the relevant legislation. In such cases, legal procedures will be initiated in addition to starting disciplinary actions against those involved.
Flexibility	Circumstances may arise during the course that prevents the instructor from fulfilling each and every component of this syllabus; therefore, the syllabus is subject to change. Students will be notified prior to any changes.